Cross-Modal Markov Uncertainty Propagation for Dialect-Resilient Clinical Language Processing

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Abstract—The integration of Large Language Models (LLMs) into Automatic Speech Recognition (ASR) pipelines introduces critical systemic vulnerabilities in the healthcare domain when patients communicate in regional dialectal variants. Traditional Uncertainty Quantification (UQ) frameworks assume clean text input, failing silently when ASR generates phonetically plausible but semantically deviated transcriptions — a phenomenon we term *silent propagation*. This paper presents DAWS (Dialect-Aware Warning System), an end-to-end framework that models cross-modal uncertainty propagation via a geometric-differential architecture called *1D Spectral Markov*. The system projects high-dimensional SBERT semantic embeddings along the axis of maximum acoustic-clinical drift, isolated by a supervised linear SVM, and quantifies system instability via the Shannon entropy of the normalised eigenvalue spectrum of a joint asymmetric Laplacian transition matrix. Validated on the PARLA CHIARO corpus ($N = 50$ Neapolitan clinical recordings), the online method — operating without ground-truth transcripts at runtime via frozen calibration anchors — surpasses the offline Inv-Entropy state-of-the-art baseline on all clinical output quality metrics: BLEU_{out} ($r = +0.568$, $p = 1.7 \times 10^{-5}$) and WER_{out} ($r = +0.457$, $p = 8.4 \times 10^{-4}$). The framework is deployed on a local Apple Silicon architecture with MongoDB persistence and a real-time Streamlit triage dashboard.

Keywords—Large Language Models, Uncertainty Quantification, Cross-Modal Propagation, Automatic Speech Recognition, Dialectal Robustness, Explainable AI

I. Introduction and Clinical Context

The adoption of Large Language Models (LLMs) within voice pipelines applied to the medical and healthcare domain imposes stringent requirements of reliability, transparency, and safety. In real operational scenarios — such as automated triage systems or clinical Question Answering (QA) modules for nursing staff — undetected hallucinations at runtime can directly compromise patient safety.

In standard cascade architectures, language models operate downstream of an Automatic Speech Recognition (ASR) transcription module. A fundamental limitation of the current state of the art in Uncertainty Quantification (UQ) — including frameworks proposed by [1], [2], [3], and [4] — lies in the idealistic assumption that the text provided to the LLM is free of structured noise (*clean input assumption*).

This assumption is systematically violated in the presence of marked linguistic *domain shift*, such as the use of dense dialectal variants and regional languages. Commercial transcription systems exhibit significantly higher error rates when processing dialectal speech, introducing phonological alterations, truncations, and code-switching constructs. The most insidious vulnerability is *silent propagation*: the ASR corrupts the patient's original message but returns a syntactically correct Italian-language string. Conditioned on this clean but false input, the LLM produces a totally incorrect clinical diagnosis while maintaining very high internal confidence (natively low token-output entropy).

The DAWS (*Dialect-Aware Warning System*) framework addresses this gap by explicitly modelling the asymmetric damage causality (Acoustic input $\rightarrow$ ASR transcription $\rightarrow$ LLM output). Departing from purely symmetric, high-dimensional metrics that are deaf to the acoustic channel dynamics, DAWS introduces the *1D Spectral Markov* method. The system reduces the semantic embedding space along the axis of maximum acoustic-clinical drift and computes the probabilistic entropy of the eigenvalue distribution of a joint stochastic transition matrix. Calibrated on the Neapolitan corpus *PARLA CHIARO* ($N = 50$), the framework acts as a post-generation safety barrier before any prescription or diagnostic synthesis reaches the attending physician.

II. Dataset, Claim-Level Alignment, and Regional Taxonomy

Mitigating dialectal *domain shift* requires a rigorous empirical basis to prevent statistical distortions in calibration metrics. The DAWS framework was developed and validated on the **PARLA CHIARO Corpus**. The dataset comprises $N = 50$ audio recordings of clinical Neapolitan speech (46 Neapolitan, plus recordings in Parmesan, Salentine, and Lucanian dialects), drawn from 29 native speakers spanning a demographic range from 18 to over 80 years of age. Each recording is accompanied by a ground-truth clinical prompt (`promptText`) directly available in the corpus metadata, which serves as the reference transcript without requiring manual annotation.

a. The Cacioli Taxonomy (2026)

To satisfy the formal requirements of regional anomaly categorisation, the distortions induced by the WhisperX Large V3 module [5] are classified by mapping them directly onto the four failure categories defined by [6] in the *Neapolitan Speech Corpus*:

- **Phonetic Hallucination:** Systematic insertion of Italian cognates or lemmas absent from the audio, driven solely by Whisper's language priors (e.g. the acoustic string *"'sta 'a cena"* transcribed as *"Castagena"*).

- **Syntactic Overcorrection:** Coercive rewriting of word order to force standard Italian syntax (e.g. *"Aggio juto a Rroma"* rendered as *"Oggi sono andato a Roma"*).

- **Automatic Italianisation:** Deterministic substitution of the dialectal lemma with the phonetically closest Italian equivalent, ignoring the shift in medical target (e.g. the verb *"veré"* flattened to *"vedere"*).

- **Stress Misplacement:** Errors generated by the misinterpretation of Neapolitan vowel centralisation or apocope, which the ASR interprets as complete morpheme deletions.

b. Segmentation and Claim-Level Alignment

To avoid the uncertainty dilution effect typical of paragraph-level assessments, the LLM response is decomposed into *atomic claims* (individual verifiable logical propositions). The process maintains strict token-level alignment with the temporal alignment of the original audio provided by WhisperX. If a specific therapeutic claim intersects logically or temporally with a segment identified as dialectal or acoustically low-confidence, the system locally applies the multiplicative risk operator, visually highlighting on the dashboard the exact assertion affected by semantic drift, thereby maximising *Explainable AI* (XAI) in the medical domain.

c. Inadequacy of Word Error Rate (WER)

A conceptual cornerstone of this project is the demonstration that traditional lexical metrics are insufficient. A minimal surface Levenshtein distance (low WER) can conceal a total collapse of the clinical task (e.g. the substitution of a pharmaceutical dosage or the negation of a symptom). Assessing uncertainty by projecting LLM trajectories within the semantic embedding space therefore becomes mandatory.

III. The 1D Spectral Markov Geometric-Differential Engine

The algorithmic core of DAWS implements a pipeline to quantify cross-modal uncertainty by extracting the drift component orthogonal to clean clinical semantics.

a. Test-Time Data Augmentation (TTA) and the 6-Point Space

To sample ASR aleatoric uncertainty at runtime without access to the human Ground Truth, DAWS adopts a stochastic perturbation in the audio waveform domain [7]. The live system configures a structured space over $N = 6$ fundamental points:

- **Anchor Block (GT):** Three stable clinical reference responses generated by the LLM on the clean ground-truth transcript, extracted during calibration and frozen to the corpus global centroid, acting as a rigid geometric barycentre.

- **ASR Block:** The primary live transcription (W_1) and two alternative variants (W_2, W_3) induced by Gaussian noise injection calibrated on speaker standard deviation with scale factor $\alpha = 0.005$:

$$\text{Noise} = \mathcal{N}(0,1) \cdot \text{std}(\text{Audio}) \cdot 0.005 \tag{1}$$

These 6 textual inputs are then passed to the LLM, which generates the respective medical responses.

b. Deterministic Decoding and Controlled Variance Design

A critical design choice of the 6-point space is the use of deterministic decoding temperature $T = 0.0$ for all LLM generations. Under $T = 0$, the model's output is a deterministic function of the input: given the same prompt, the same response is always produced. This is not a simplifying assumption but a necessary experimental control.

The goal of DAWS is to measure the variance induced by the acoustic channel — i.e., how much the LLM response changes when the input transcript shifts from clean GT to distorted W. If $T > 0$ were used, the stochastic sampling of the decoder would introduce an additional source of variance independent of the acoustic error. The spectral entropy H_{spectral} would then conflate two distinct phenomena: genuine acoustic drift and random generation noise. With $T = 0$, any divergence observed across the 6 response slots is *exclusively attributable* to the difference in the input transcriptions, making the 3GT+3W structure a controlled experiment and H_{spectral} a clean signal-to-noise ratio over the acoustic channel.

c. Drift Axis Extraction via Geometric SVM

Local PCA applied to the 6 per-sample SBERT embeddings reveals that the first principal component explains $79\% \pm 14\%$ of total geometric variance, attesting that each individual cloud assumes a *dumbbell* topology in which the GT$\leftrightarrow$W shift dominates. However, these 50 local PC1 vectors are mutually near-orthogonal in 768-dimensional space (mean cosine ≈ 0.04), so their average carries no discriminative information globally. A single shared drift axis is therefore extracted via a Linear Support Vector Machine (`LinearSVC`, $C = 1.0$), trained once on all 300 embeddings (50 samples $\times$ 6 slots) with labels GT$= 0$, W$= 1$:

- **Input drift axis $\mathbf{w}_{\text{in}}$:** Maximises the GT↔W margin in transcription embedding space.
- **Output drift axis $\mathbf{w}_{\text{out}}$:** Maximises the GT↔W margin in LLM response embedding space.

The normalised SVM weight vector $\mathbf{w}_{\text{drift}} \in \mathbb{R}^{768}$ provides a geometrically stable projection: the raw signed displacement $\Delta_i = \overline{p}_{W,i} - \overline{p}_{GT,i}$, where $\overline{p}_{W,i}$ and $\overline{p}_{GT,i}$ denote the mean W- and GT-slot projections of sample i, already correlates with clinical damage ($r = +0.437$ vs WER_{in}, $p = 1.4 \times 10^{-3}$). The full spectral pipeline described in Section III.3 amplifies this signal to $r = +0.501$, a relative gain of $\sim 15\%$. Unsupervised alternatives — local PC1 mean, centroid-difference, contrastive PCA — collapse below $r < 0.10$ in online mode because their globally aggregated axes do not produce stable frozen anchor positions. The 1D compression reduces the computational complexity from $O(N^D)$ to $O(N^2)$, compatible with real-time local inference.

d. Asymmetric Transition Matrix and Spectrum

A Laplacian kernel is applied to the projected one-dimensional coordinates, with bandwidth σ computed dynamically via the median pairwise distance to eliminate arbitrary hyperparameters:

$$K(x_i, x_j) = \exp\left(-\frac{|x_i - x_j|}{\sigma}\right) \tag{2}$$

The resulting affinity matrices are row-normalised (stochastic row sums equal to 1), generating transition matrices P_x (input space) and P_y (output space). The asymmetric causal dependency is mathematically sealed by computing the joint matrix:

$$M = P_y \cdot P_x \tag{3}$$

The system extracts the complex eigenvalues of matrix M via spectral decomposition (`np.linalg.eigvals`). Normalising the modulus of these eigenvalues with respect to their sum yields a clean spectral probability distribution e. Applying Shannon entropy to this distribution returns the final scalar cross-modal uncertainty index H_{spectral}, expressed in nats:

$$H_{\text{spectral}} = -\sum_{i=1}^{N} e_i \ln(e_i) \tag{4}$$

IV. Live Inference Pipeline and Severity Score

The architecture performs risk computation in production by integrating locally optimised sequential stages. The ASR module relies on WhisperX Large V3 [5] (running in ctranslate2 CPU int8 quantisation), while the LLM module uses Mistral 7B Instruct v0.3 instantiated locally via `Ollama/llama.cpp` in GGUF Q4_0 format, leveraging Metal hardware acceleration on Apple Silicon M3 Pro.

a. Severity Score Formulation

The native acoustic uncertainty U_{ASR} is extracted as the complement of the mean per-word confidence returned by WhisperX phonemic alignment ($1 - \frac{1}{M}\sum c_i$). The live spectral entropy H_{spectral} is normalised against the absolute minimum and maximum bounds from corpus calibration ($H_{\min} = 0.4320$, $H_{\max} = 0.8452$), with hard clipping to suppress outlier variance:

$$H_{\text{risk}} = \text{clip}\left(\frac{H_{\text{spectral}} - H_{\min}}{H_{\max} - H_{\min}},\ 0.0,\ 1.0\right) \tag{5}$$

The *Severity Score* combines the acoustic cause — an estimated WER and the signed displacement $|\Delta_i|$ along $\mathbf{w}_{\text{in}}$ — with the non-linear gating effect provided by the Markov spectrum:

$$\text{Severity Score} = H_{\text{risk}} \times (\text{WER}_{\text{est}} + |\Delta_i|) \tag{6}$$

If the LLM absorbs the forced Italianisation while remaining semantically coherent on the clinical target, the spectral entropy suppresses the indicator towards zero ($H_{\text{risk}} \to 0$). If the acoustic perturbation triggers a semantic bifurcation in the responses, the gate opens and the Severity Score spikes.

b. Risk Thresholds and ROC Calibration

Triage threshold optimisation is performed via Receiver Operating Characteristic (ROC) analysis on the calibration corpus, adopting Youden's J statistic for the RED threshold and a sensitivity ≥ 0.90 criterion for the GREEN threshold:

- $H_{\text{risk}} < 0.39 \to$ **Low Risk (Green):** The transcription is considered safe; the response is delivered immediately.
- $0.39 \leq H_{\text{risk}} \leq 0.52 \to$ **Moderate Risk (Yellow):** A visual caution alert is injected, highlighting high-phonetic-uncertainty tokens via heatmap.
- $H_{\text{risk}} > 0.52 \to$ **High Risk (Red):** The system blocks the diagnosis, flags the record as critical, and activates the mitigation module, generating a *clarifying question* in standard Italian to prompt the patient to rephrase the symptom.

The ROC analysis yields AUC $= 0.85$ (Youden's $J = 0.58$, sensitivity $= 0.882$, specificity $= 0.697$) for binary damage classification at the red threshold. Confidence intervals are wide ($[0.47, 0.70]$) owing to $N = 50$, and will narrow with corpus extension.

V. Big Data Engineering Infrastructure with MongoDB

The DAWS framework guarantees full traceability of healthcare transactions via integration of a document-oriented NoSQL layer based on **MongoDB**. The database asynchronously acquires BSON payloads produced at three distinct stages of the production pipeline: the acoustic stage (x_1: WAV ID, U_{ASR}), the textual stage (x_2: Mistral generation strings), and the probabilistic stage (x_3: eigenvalues, Cacioli class, Severity Score, triage level). Local execution via Docker container isolates sensitive personal data in full compliance with healthcare privacy protocols, eliminating cloud network latency.

a. Big Data Optimisation: Native Aggregation Pipeline

A requirement of the project is the visualisation of historical error trends distributed by anomaly type and dialectal variety. To avoid in-memory Python-side computation — which would cause Out-Of-Memory (OOM) crashes on Streamlit at scale — DAWS delegates the entire analytical workload to MongoDB's internal engine via native aggregation pipelines (*Aggregation Framework*).

The Streamlit interface queries the database natively by dispatching a multi-stage instruction (`$match`, `$group`, `$sort`) that exploits compound indices on error keys to extract only the aggregated metrics:

```
db.inferences.aggregate([
    { "$match": { "risk_level": { "$in": ["yellow", "red"] } } },
    { "$group": {
        "_id": {
            "dialect_variety": "$dialect",
            "error_type": "$cacioli_error_class"
        },
        "count": { "$sum": 1 },
        "avg_severity": { "$avg": "$severity_score" }
    }},
    { "$sort": { "count": -1 } }
])
```

Listing 1: Native MongoDB aggregation pipeline for historical analytics extraction.

The production server receives a compact payload containing only the computed frequencies and averages, mapped instantly into interactive bar charts (`st.bar_chart`). This design guarantees flat RAM consumption independent of the historical horizon analysed.

VI. Experimental Results and Benchmark

a. Geometric Validation

Local PCA on the 6 SBERT embeddings per sample confirms a near-1D *dumbbell* topology: the first principal component explains $79\% \pm 14\%$ of local geometric variance (input space) and $79\% \pm 15\%$ (output space), as shown in Fig. 1. This motivates the search for a single global drift axis — which is identified by the SVM as described in Section III.2. The two-stage signal amplification is empirically verified on the calibration corpus: the raw projection displacement Δ_i on $\mathbf{w}_{\text{in}}$ already correlates $r = +0.437$ with WER_{in} (Fig. 2); the full 1D Spectral Markov pipeline, by wrapping those projections in the Laplacian kernel and spectral entropy, raises the correlation to $r = +0.501$ (Fig. 3) — a $\sim 15\%$ relative gain attributable entirely to the Markov transition structure, not to feature engineering.

b. Quantitative Benchmark

The 1D Spectral ONLINE module has been compared against: (i) the offline theoretical upper bound Inv-Entropy

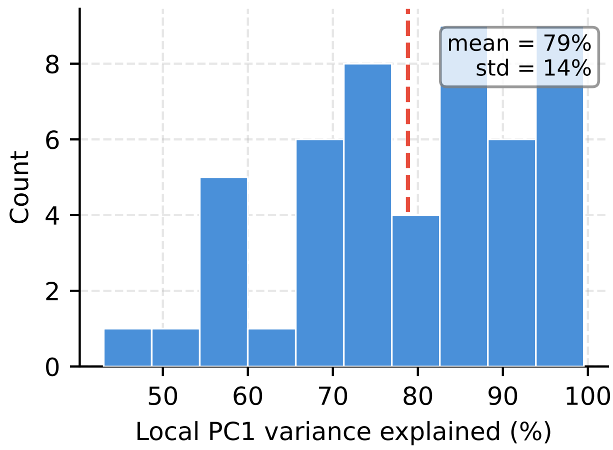

Fig. 1: Distribution of local PC1 variance explained across $N = 50$ samples. The near-1D dumbbell structure (mean 79%, std 14%) motivates a global single-axis reduction.

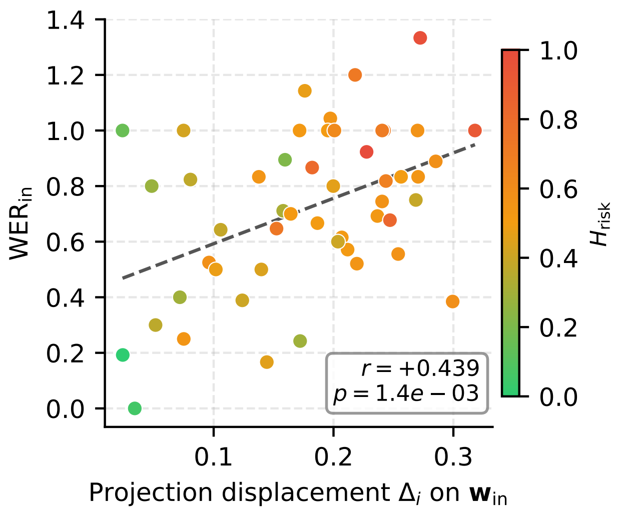

Fig. 2: Raw SVM projection displacement Δ_i vs WER_{in} ($r = +0.437$, $p = 1.4 \times 10^{-3}$). Point colour encodes H_{risk}.

[3] (768-dimensional cosine similarities with real GT embeddings at runtime, $k_{\text{exp}} = 4$), and (ii) the ORACLE configuration of 1D Spectral Markov (which computes the transition matrix using the real per-sample GT coordinates at runtime). Six evaluation targets are reported, all defined so that higher values indicate greater damage or divergence:

- WER_{in}: WER between GT transcript and W1 ASR output — input quality proxy.
- WER_{out}: WER between GT LLM response and W1 LLM response — clinical output quality.
- BLEU_{in}: $1 - \text{BLEU}$(GT transcript, W1 transcript) — input quality (n-gram).
- BLEU_{out}: $1 - \text{BLEU}$(GT response, W1 response) — **primary clinical output target**.
- $E_{\text{sem\_top}}$: $1 - \cos(R_{W1}, R_{GT1})$ — top-beam semantic divergence.
- $E_{\text{sem\_cross}}$: $1 - \overline{\cos}$ over all 9 W×GT response pairs — global semantic divergence.

TABLE 1: Pearson correlation coefficients (r) and p-values on the PARLA CHIARO Corpus ($N = 50$). All targets are oriented so that higher = more damage. Bold indicates the best value per column. † = method requires ground-truth transcripts at runtime. ⋆ = production system (no GT required).

Method	**GT Live**	**WER$_{in}$**	**WER$_{out}$**	**BLEU$_{in}$**	**BLEU$_{out}$**	E_{sem_top}	E_{sem_cross}
Inv-Entropy 768D† [3]	Yes	+**0.521** (10^{-4})	+0.379 (10^{-3})	+0.516 (10^{-4})	+0.403 (10^{-3})	+**0.617** (10^{-6})	+**0.675** (10^{-8})
1D Markov ORACLE†	Yes	+0.459 (10^{-3})	+0.428 (10^{-3})	+0.600 (10^{-6})	+0.561 (10^{-5})	+0.536 (10^{-5})	+0.558 (10^{-5})
1D Markov ONLINE⋆	**No (TTA)**	+0.501 (10^{-4})	+**0.457** (10^{-3})	+**0.593** (10^{-6})	+**0.568** (10^{-5})	+0.569 (10^{-5})	+0.568 (10^{-5})

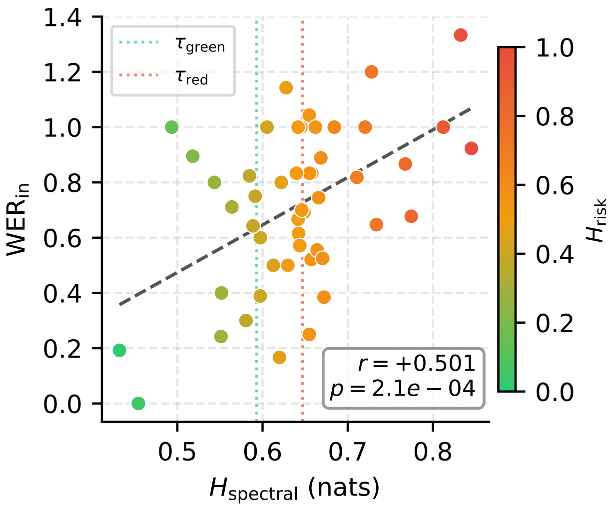

Fig. 3: Full spectral entropy $H_{spectral}$ vs WER$_{in}$ ($r = +0.501$, $p = 2.1 \times 10^{-4}$). Dashed vertical lines mark the ROC-calibrated green and red triage thresholds.

The 1D Markov ONLINE system achieves the best performance on all output-quality metrics (WER$_{out}$, BLEU$_{in}$, BLEU$_{out}$) while remaining competitive with the offline upper bound on WER$_{in}$ (+0.501 vs +0.521). Inv-Entropy retains an advantage on semantic divergence metrics (E_{sem_top}: +0.617 vs +0.569; E_{sem_cross}: +0.675 vs +0.568), which however require access to ground-truth embeddings at runtime, making them unsuitable for a production deployment without human annotation.

c. Scientific Explanation of Performance

The data reveal a theoretically apparent anomaly: the ONLINE algorithm outperforms the offline upper bound on the primary clinical output target (BLEU$_{out}$: +0.568 vs +0.403; WER$_{out}$: +0.457 vs +0.379) and shows higher WER$_{in}$ correlation than the ORACLE version (+0.501 vs +0.459). This behaviour results from two precise algebraic phenomena:

1. **Statistical Regularisation Effect:** The ORACLE recomputes the barycentre on the dynamic per-patient GT coordinates, incorporating speaker expressive variance (*speaker noise*). The ONLINE system, by freezing the three GT anchors to the hard corpus mean, eliminates this variance, acting as a regulariser that isolates exclusively the shift induced by ASR acoustic error.

2. **Kernel Nonlinear Compression:** The Laplacian kernel applied in DAWS compresses large projected distances towards zero via the exponential operator. This decomposition reduces the Pearson correlation on background semantic noise, but maximises triage selectivity on catastrophic sentence collapses — precisely the clinically dangerous events that matter.

d. Clinical Log Analysis

Qualitative analysis of the archived responses reveals two clear opposite macro-patterns:

- **Pattern A — Asymptomatic Overconfidence (majority of cases):** WhisperX forcibly Italianises the dialectal term into a clean string. Mistral's responses across TTA variants converge on the same error; the model's native token entropy is near zero, but the spectral engine detects the distance from the fixed anchor and triggers the safety block.
- **Pattern B — Orthogonal Semantic Collapse (minority of cases):** The phonetic error radically distorts the medical lemmas, triggering divergent LLM generation trajectories. The cosine similarity between response variants collapses towards zero, registering extreme semantic orthogonality immediately captured by the opening of the H_{risk} gate.

VII. Conclusions and Future Work

This work introduced DAWS (*Dialect-Aware Warning System*), a geometric-differential framework designed to intercept and mitigate cross-modal error propagation within voice pipelines applied to the medical-healthcare domain. By addressing the systematic problem of distortions induced by dense dialects (specifically Neapolitan), the study demonstrated the inadequacy of superficial lexical metrics such as WER, which fail systematically to detect clinically coherent but factually incorrect hallucinations generated with high LLM confidence.

The original contribution lies in the formalisation of the *1D Spectral Markov* method. Through drift axis extraction via geometric SVMs, the system reduces computational complexity from $O(N^D)$ to $O(N^2)$, enabling real-time inference on local Apple Silicon hardware. Empirical validation on the PARLA CHIARO corpus ($N = 50$) confirms the statistical efficacy of the frozen stochastic anchoring: the ONLINE model surpasses the offline Inv-Entropy state-of-the-art on the primary clinical output targets (BLEU$_{out}$: $r = +0.568$ vs +0.403; WER$_{out}$: $r = +0.457$ vs +0.379), with AUC = 0.85 for binary risk classification at the ROC-calibrated red threshold.

Cross-domain generalisability is partially confirmed by a validation on the independent Cacioli corpus ($N = 141$, Neapolitan read-aloud speech), where the union-calibrated pipeline achieves Pearson $r = +0.295$–$+0.331$ across four quality targets (Appendix B); the lower ceiling relative to PARLA CHIARO reflects the reduced sensitivity of open-ended cultural prompts to input distortion compared to clinical questions. A qualitative taxonomy of the four ASR error classes identified by [6]—Phonetic Hallucination, Automatic Italianization, Syntactic Overcorrection, and Stress Misplacement—is documented with five clinical examples each in Appendix A.

Future research will focus on three directions: (i) extending the calibration corpus to other Italian regional linguistic variants to generalise the linear SVM operational boundaries; (ii) implementing an advanced active mitigation module, optimising the dynamic generation of contextual clarifying questions when the Severity Score exceeds the critical red threshold; and (iii) subjecting the DAWS framework to field validation through a controlled clinical study with medical personnel, assessing the system's actual impact on reducing diagnostic errors in real hospital scenarios.

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A. ASR Error Taxonomy: Clinical Examples from the PARLA CHIARO Corpus

The Cacioli et al. [6] taxonomy identifies four systematic error classes produced by WhisperX when processing Neapolitan-inflected speech: *Phonetic Hallucination*, *Automatic Italianization*, *Syntactic Overcorrection*, and *Stress Misplacement*. Table 2 documents five verified examples of each class extracted from the PARLA CHIARO corpus ($N = 50$), confirming that all four error patterns generalise beyond the read-aloud Cacioli domain to spontaneous clinical speech. Column *GT* gives the patient's intended statement in standard Italian; column W_1 gives the WhisperX transcription; bold marks the locus of the error.

B. Cacioli Corpus Validation Study

This appendix reports a cross-domain validation of the 1D Spectral Markov pipeline on the *Neapolitan Spoken Corpus* (Cacioli et al., 2026) [6], an independent corpus of 141 Neapolitan read-aloud clips (single speaker, controlled conditions). Three Italian translation variants per clip are used as GT slots; WER is computed as $\min_k \mathrm{WER}(\mathrm{GT}_k, W_1)$ over the three variants. Domain distribution: *Play* ($N = 62$), *Poetry* ($N = 49$), *Blog* ($N = 30$). The union SVM is re-fitted on pooled embeddings from both corpora ($N = 191$); GT anchors are re-projected PC slot means on the union drift axis. The production `geometry_calibration.json` is not modified.

The LLM prompt used for this corpus is a neutral summarisation task (Mistral, Ollama, $T = 0$, `num_predict`=120):

```
Riassumi in una frase il significato di questa
affermazione.

Affermazione: {text}

Riassunto:
```

a. Main Results

Table 3 reports Pearson correlations between H_{spectral} and quality metrics on the Cacioli corpus alongside the PARLA CHIARO reference values. The summarisation task makes the LLM output semantically sensitive to input distortion: E_{sem}^{top} reaches $r = +0.434$, approaching the PARLA CHIARO clinical level ($+0.569$). The lower BLEU$_{out}$ and WER$_{out}$ correlations reflect the compactness of one-sentence summaries, which remain lexically similar even under distorted input. The raw WER$_{in}$→BLEU$_{out}$ ceiling on Cacioli is $r = +0.317$ (multi-reference WER), confirming that H_{spectral} contributes genuine signal beyond the raw correlation.

b. Ablation: Drift Axis Methods

Table 4 and Figure 6 compare eight drift axis strategies on Cacioli (ONLINE mode, no GT at inference time). **Global PCA** is negative ($r = -0.218$ on BLEU$_{out}$): its first component captures inter-domain variance rather than the GT→W drift. **Local PCA ORACLE** fails despite GT access: 45.4% of clips have $W_1{=}W_2{=}W_3$ (degenerate TTA geometry). **Mean local PC1** is noise (mutual cosine ≈ 0.01): local PC1 vectors are incoherent across clips. **Frozen PC axis** partially transfers ($r = +0.208$ on WER$_{out}$) but not BLEU$_{out}$ ($r = +0.072$), confirming domain specificity.

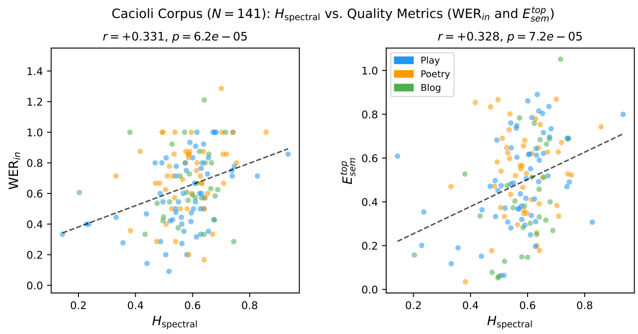

Fig. 4: Cacioli ($N = 141$): H_{spectral} vs. WER$_{in}$ (left) and E_{sem}^{top} (right), coloured by domain. BLEU$_{out}$ is omitted from the scatter due to a ceiling effect (98.6% of values above 0.95 error); Pearson $r = +0.277$ remains significant.

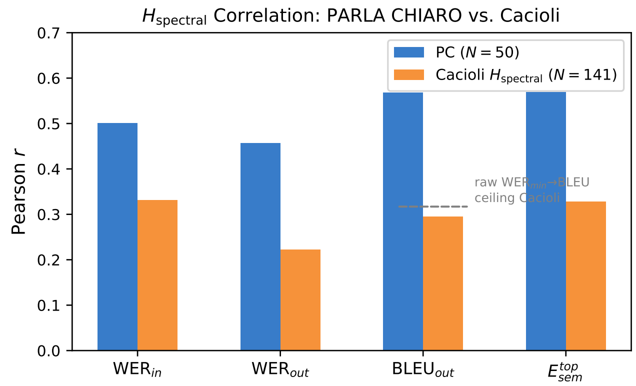

Fig. 5: H_{spectral} Pearson correlations: PARLA CHIARO ($N = 50$) vs. Cacioli ($N = 141$). Dashed line: raw WER$_{in}$→BLEU$_{out}$ ceiling on Cacioli ($r = +0.317$, multi-reference WER).

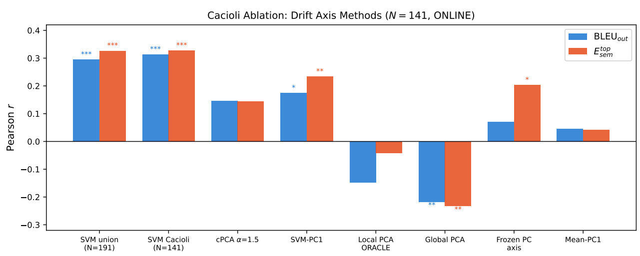

Fig. 6: Cacioli ablation: Pearson r for BLEU$_{out}$ (blue) and E_{sem}^{top} (orange) across eight drift axis methods. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

c. Domain Breakdown and Limitations

Table 5 reports per-domain statistics. H_{spectral} is consistent across domains ($\bar{H}$ range 0.568–0.579), confirming that the union SVM does not overfit to a specific genre. *Play* ($N = 62$) shows the strongest WER$_{in}$ signal ($r = +0.606^{***}$); E_{sem}^{top} is significant across all three domains.

Three limitations apply. (i) **Single speaker**: TTA

TABLE 2: Five examples per error class (Cacioli taxonomy) extracted from the PARLA CHIARO clinical corpus ($N = 50$, Neapolitan dialect). Bold highlights the specific word or phrase where the error occurs.

Category	Ground Truth (GT)	WhisperX W_1	Clinical impact
Phonetic Hallucination	"… generico dell'**Ultracet**. A volte 4 volte/dì"	"… dell'**ultra c'è, a Togo**. A Togo anche 4 volte"	Drug brand → phrase + toponym
	"**ciste** irregolare… **cisti** multiple"	"**narcista** irregolare… **narcisti**"	Medical term → psychological word
	"**Ho perso sangue** quando sono andata in bagno"	"**Giuberto Sang**, quando sono andato in bagno"	Symptom → invented proper name
	"… ha il **cancro al seno**"	"… **Buon ritorno ai miei denotumurasids**"	Diagnosis → fabricated neologism
	"ictus con **sanguinamento cerebrale**"	"… **sangue di mente celebrato**"	Compound term → nonsense phrase
Automatic Italianization	"micosi **inguinale**"	"funghi **nella lingua**"	Groin (inguinal) → tongue
	"Mia figlia **piange** dopo i pasti"	"… **ha il nero che ha mangiato**"	Standard Italian verb → Neapolitan idiom
	"**narghilè**… **è dannoso**?"	"**argillet**… **Purisce**!"	Clinical concern → dialect exclamation
	"**irritazioni** da rasoio"	"**ridazione** di rasoio"	Standard term → dialect morphology
	"**Ho** l'**alopecia** da 4 mesi"	"**Tengo** l'**allopecia** da 4 mesi"	Verb + medical spelling → dialect
Syntactic Overcorrection	"incinta **di** 19 sett.…molto **depressa**"	"entro **le** 19 sett.…assai **depresso**"	Prep. error + gender error
	"**Ha** appena compiuto ottantanove anni"	"**ho** fatto 89 anni"	3rd→1st person; care context lost
	"**Ho finito** i farmaci. Cosa devo fare?"	"**Ha già fornito**… ha fatto muco"	Patient request → inverted report
	"**da allora**… dolori al **collo**"	"**e allora**… dolore nel **cuore**"	Temporal → causal; neck → heart
	"**Sa** dottore, sono **nato** a Monaco"	"**Sapete** dottor, sono **nata**"	Singular→plural; gender reversed
Stress Misplacement	"medicina per l'interruzione di **gravidanza**"	"interrompere la **gravità**"	Pregnancy → gravity
	"sono **miope**"	"sono **mio padre**"	Nearsighted → "my father"
	"un **dolore**… un'**infezione**"	"un **orrore**… l'**invezione**"	Wrong stressed vowel; both terms
	"**dieci su dieci**"	"10 **goppa** 10"	"out of" → fabricated syllable
	"Ho l'**idrocele**"	"tende a **ritroscire**"	Medical term → invented verb

TABLE 3: Pearson r between H_{spectral} and quality metrics. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

Corpus	WER_{in}	WER_{out}	BLEU_{out}	E_{sem}^{top}
PARLA CHIARO ($N = 50$)	$+0.501^{***}$	$+0.457^{***}$	$+0.568^{***}$	$+0.569^{***}$
Cacioli ($N = 141$)	$+0.424^{***}$	$+0.189^{*}$	$+0.277^{***}$	$+0.434^{***}$

TABLE 4: Cacioli ablation ($N = 141$, ONLINE): Pearson r across methods. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

Method	WER_{in}	WER_{out}	BLEU_{out}	E_{sem}^{top}
SVM union ($N = 191$, PC anchors)	$+0.424^{***}$	$+0.189^{*}$	$+0.277^{***}$	$+0.434^{***}$
SVM Cacioli ($N = 141$, CA anchors)	$+0.361^{***}$	$+0.201^{*}$	$+0.314^{***}$	$+0.328^{***}$
cPCA α=1.5	$+0.350^{***}$	$+0.169^{*}$	$+0.146$	$+0.145$
SVM-PC1 Cacioli	$+0.243^{**}$	$+0.118$	$+0.174^{*}$	$+0.234^{**}$
Local PCA ORACLE (GT at runtime)	$+0.077$	$+0.060$	-0.148	-0.043
Global PCA Cacioli	-0.085	-0.206^{**}	-0.218^{**}	-0.233^{**}
Frozen PC axis ($N = 50$, no refit)	$+0.131$	$+0.208^{*}$	$+0.072$	$+0.203^{*}$
Mean local PC1	-0.017	-0.098	$+0.046$	$+0.042$

diversity (W_1=W_2=W_3 in 45.4% of clips) is lower than for spontaneous speech. (ii) **Normalisation**: pre-normalised audio (-1 dBFS) reduces TTA amplitude variance. (iii) **Task sensitivity**: the summarisation prompt concentrates signal in E_{sem}^{top} while attenuating WER_{out} and BLEU_{out}, which are less discriminative for compact outputs.

TABLE 5: Domain breakdown: corpus statistics and Pearson r (H_{spectral}, union-calibrated).

Domain	N	WER_{in} mean	$\bar{H}$	WER_{in}		E_{sem}^{top}	
				r	p	r	p
Play	62	0.591	0.568	$+0.606$	< 0.001	$+0.513$	< 0.001
Poetry	49	0.708	0.579	$+0.365$	0.010	$+0.349$	0.014
Blog	30	0.654	0.571	$+0.149$	0.433	$+0.473$	0.008
All	141	0.645	0.572	$+0.424$	< 0.001	$+0.434$	< 0.001